








Blockchain Applications

Lecture 2: Stylized Facts, Efficiency and Bubbles



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Today's Question and Route

- **Question**: how do crypto prices behave, and what does that mean for a business that holds, accepts or prices crypto?
  - a treasurer, an exchange or a fund needs numbers, not narratives
- **Route**
  - from prices to returns on a 7-day market
  - the shape of returns: heavy tails, rare but frequent crashes
  - volatility through time, and the link with equities and gold
  - is the market efficient? can past prices predict returns?
  - bubbles, drawdowns and network value (Metcalfe's law)
- Next lecture: models for volatility and tail risk: generalised autoregressive conditional heteroskedasticity (GARCH), value at risk (VaR), expected shortfall (ES)

Learning Outcomes

- Compute log returns and annualise volatility with the right number of days
- Read a summary table: volatility, skewness, kurtosis, worst day
- Use a histogram and a quantile–quantile (QQ) plot to show why the Normal distribution understates crash risk
- Interpret rolling volatility and rolling correlation charts
- Explain weak-form efficiency and read a variance-ratio test
- Measure drawdowns and judge bubble and network-value claims with data

Abbreviations

ACF autocorrelation function
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CI confidence interval
COVID-19 coronavirus disease 2019
CRIX CRypto IndeX (Trimborn & Härdle (2018))
EODHD EOD (end-of-day) Historical Data, data vendor
ES expected shortfall
EWMA exponentially weighted moving average
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FTX crypto exchange (company name), bankrupt in November 2022
GARCH generalised autoregressive conditional heteroskedasticity
i.i.d. independent and identically distributed
IDA Institute Digital Assets
JB Jarque–Bera (test statistic)
LPPL log-periodic power law
LPPLS log-periodic power law singularity
MBB Master Blockchain for Business
QQ quantile–quantile
S&P 500 Standard & Poor's 500 US stock index
sd standard deviation
SE standard error
SFM Statistics of Financial Markets (master course; Quantlet prefix)
USD US dollar
UTC Coordinated Universal Time
VaR value at risk
VR variance ratio

Reading and Tools

- Core reading: Trimborn & Härdle (2018), *CRIX: an index for cryptocurrencies*; Peterson (2018), *Metcalfe's law as a model for Bitcoin's value*
- Video units on Quantinar (course Blockchain Analytics)
  - Blockchain Mechanism and Distributional Characteristics of Cryptos
  - Crypto Currency Returns
- Data
  - daily closes of Bitcoin, Ethereum, the S&P 500 and gold, 2018 – 18 September 2026 (EODHD)
  - on-chain transactions of Bitcoin, 2014–2025 (ETH Chronicles project, Lecture 1)
- Further: Pele et al. (2023), *Are cryptos becoming alternative assets?*

Log Returns and Annualisation

- Daily **log return** in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; P_t: the close on day t
- Annual volatility from daily data: $\sigma_a = s\sqrt{A}$
  - s: the standard deviation of daily returns; A: the number of returns per year
  - crypto trades every day: $A = 365$; stocks: $A \approx 252$ trading days
- **Worked example**: Bitcoin 2018–2026, $s = 3.355\%$ a day
  - right: $\sqrt{365} \times 3.355 = 19.10 \times 3.355 = 64.1\%$ a year
  - wrong: $\sqrt{252} \times 3.355 = 53.3\%$, an understatement of 17%
- Rule: annualise each asset with its own calendar; compare assets only after that

Moments of the Return Distribution

- For n daily returns $r_1, \dots, r_n$:

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad s^2 = \frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2$$

$$\hat{S} = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t - \bar{r}}{s}\right)^3, \qquad \hat{K} = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t - \bar{r}}{s}\right)^4 - 3$$

  - $\bar{r}$: mean; s: standard deviation (volatility); $\hat{S}$: skewness; $\hat{K}$: excess kurtosis
  - Normal distribution: $S = 0$ and $K = 0$

- **Jarque–Bera (JB) test** of normality:

$$\mathrm{JB} = \frac{n}{6}\left(\hat{S}^2 + \frac{\hat{K}^2}{4}\right) \sim \chi^2_2 \text{ under } H_0\text{: Normal}$$

  - reject normality at 5% if $\mathrm{JB} > 5.99$
  - 2018–2026: Bitcoin 29 620, Ethereum 16 593, S&P 500 20 462, gold 13 203: normality is rejected for all four

Worked Example: Jarque–Bera for Bitcoin

- Bitcoin, 2018–2026: $n = 3\,183$, $\hat{S} = -0.945$, $\hat{K} = 14.825$

$$\mathrm{JB} = \frac{3\,183}{6}\left(0.945^2 + \frac{14.825^2}{4}\right) = 530.5 \times (0.893 + 54.945) = 29\,622$$

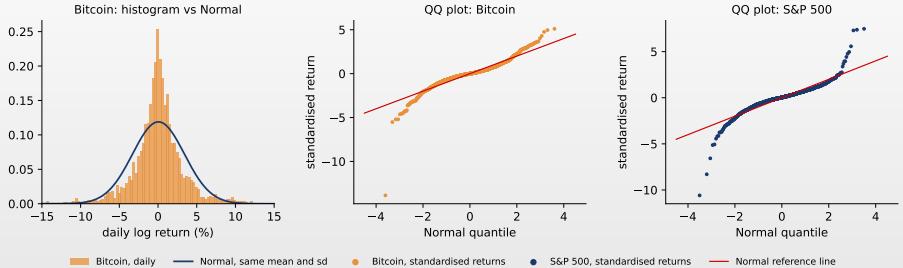
- 5% critical value of χ^2_2: 5.99; p-value $< 10^{-300}$
- Almost all of the statistic comes from kurtosis (54.9 of 55.8): the problem is the tails, not the asymmetry
- Caution: with n in the thousands, JB rejects even mild deviations; the size of $\hat{K}$ matters more than the p-value

Four Assets Side by Side, 2018–2026

	n	sd (%)	ann. vol. (%)	skewness	excess kurtosis	worst day (%)	days $\lvert r\rvert > 5\%$ (%)
Bitcoin	3 183	3.36	64.1	−0.95	14.8	−46.5	10.7
Ethereum	3 183	4.41	84.3	−0.83	11.1	−55.1	17.8
S&P 500	2 189	1.21	19.2	−0.65	14.9	−12.8	0.6
Gold	2 633	0.99	15.7	−0.76	10.9	−11.1	0.2

- Daily log returns, 1 January 2018 – 18 September 2026, each asset on its own calendar
  - **skewness**: 0 for a symmetric distribution; negative: the left tail (losses) is longer
  - **excess kurtosis**: 0 for the Normal distribution; positive: more extreme days than the Normal predicts
- Interpretation
  - Bitcoin is 3.3 times as volatile as the S&P 500; it moves by more than 5% on one day in nine
  - the worst day of Bitcoin and Ethereum was the same: 12 March 2020 (COVID-19 crash)
  - all four are skewed to the left and heavy-tailed

The Normal Distribution Does Not Fit

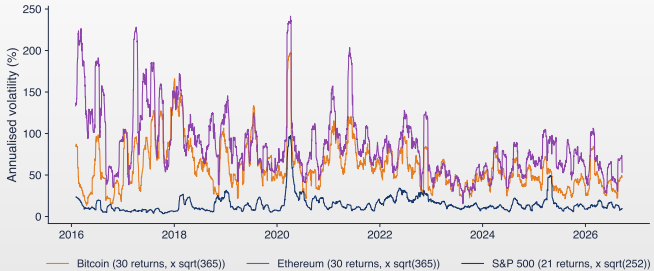


- Left: too many quiet days in the centre and too many extreme days in the tails, too few in between
- **QQ plot**: sorted returns against the quantiles a Normal distribution would give; Normal data lie on the red line
- Both QQ plots bend away at the ends: heavy tails in crypto *and* in equities

How Rare Is a Crash?

- Standardise: $z_t = (r_t - \bar{r})/s$, the number of standard deviations of each day
- Under the Normal distribution
  - $P(|z| > 3) = 0.27\%$; $P(|z| > 4) = 0.0063\%$
  - in 3 183 days we would expect 0.2 days beyond 4 standard deviations
- Bitcoin, 2018–2026: **17 days** beyond 4 standard deviations
- Worst day, 12 March 2020: -46.5%, i.e. $z = -13.9$
  - under the Normal: $P(z \leq -13.9) \approx 5 \times 10^{-44}$, i.e. once in 2×10^{43} days
- Business lesson: a risk model built on the Normal distribution will be surprised many times; the next lecture fixes this

Volatility Through Time



- Volatility on about one month: 30 daily returns $\times\sqrt{365}$ for crypto, 21 returns $\times\sqrt{252}$ for the S&P 500
- Medians: 54% (Bitcoin), 77% (Ethereum), 12% (S&P 500); all three peak in March–April 2020
- Bitcoin's volatility trends down, but it was below that of the S&P 500 on only 0.1% of the days

SFM_ch14_asset_class

Autocorrelation and the Ljung–Box Test

- **Sample autocorrelation** at lag k:

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^{n}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

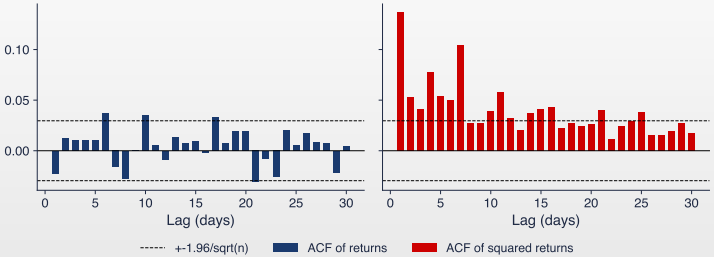
  - under independence, $\hat{\rho}(k)$ lies within $\pm 1.96/\sqrt{n}$ with probability 95%
  - autocorrelation function (ACF) of r_t: is the direction predictable? ACF of r_t^2: is the size of the move predictable?

- **Ljung–Box** test of the first m autocorrelations together:

$$Q(m) = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}(k)^2}{n-k} \sim \chi^2_m \text{ under } H_0\text{: no autocorrelation}$$

  - Bitcoin, $m = 10$: $Q = 20.2$ for r_t and $Q = 212.7$ for r_t^2; the 5% critical value is 18.3
  - weak dependence in returns, strong dependence in squared returns

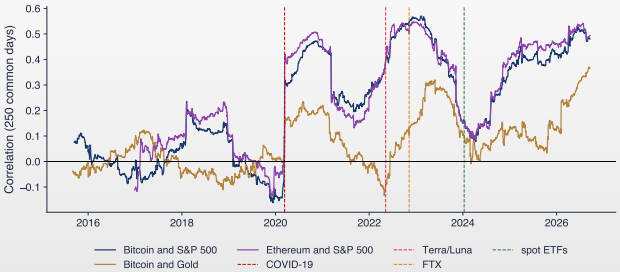
Volatility Clusters



- **Autocorrelation** at lag k: the correlation between today's value and the value k days ago; dashed: the 95% band of pure noise
- Returns (left): almost no autocorrelation; the *direction* of tomorrow's move is hard to predict
- Squared returns (right): clearly positive; the *size* of tomorrow's move is predictable: calm days follow calm days, turbulent days follow turbulent days
- This is **volatility clustering**, the basis of the GARCH models of Lecture 3

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Does Bitcoin Move with Equities?



- Correlation of daily returns on 250 common days; prices are joined on common days first, then returns are computed
- Before 2020: no link with the S&P 500 (0.02); after: 0.39 on average, peak 0.57 in February 2023
- Since 2020 crypto behaves as a **risk asset**: it falls with equities when interest rates rise (2022)

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Weak-Form Efficiency and the Variance Ratio

- Fama (1970): in a **weak-form efficient** market, past prices do not help to predict future returns
  - if they did, traders would exploit the pattern until it disappeared
- **Variance ratio** (VR) Lo & MacKinlay (1988): compare the variance of q-day returns with q times the daily variance

$$\mathrm{VR}(q) = \frac{\mathrm{Var}(r_t + r_{t-1} + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}$$

  - random walk (no predictability): $\mathrm{VR} = 1$
  - $\mathrm{VR} > 1$: trends (momentum); $\mathrm{VR} < 1$: reversals (mean reversion)
  - a test statistic Z^*, robust to volatility clustering, rejects the random walk at 5% if $|Z^*| > 1.96$

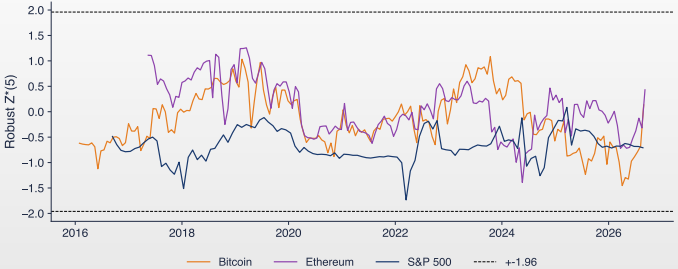
Variance-Ratio Tests

	VR(2)	VR(5)	VR(10)	$Z^*(5)$ (p-value)
Bitcoin	0.977	0.991	1.027	−0.19 (0.853)
Ethereum	0.988	1.067	1.137	1.17 (0.241)
S&P 500	0.875	0.849	0.813	−1.28 (0.201)

- Daily log returns since 2014 (Ethereum since 2016) to 18 September 2026
- Interpretation
  - all variance ratios are close to 1; no test rejects the random walk
  - the same holds in each sub-period 2014–2017, 2018–2021, 2022–2026
  - past prices alone do not give a reliable trading rule

SFM_ch14_garch_efficiency

Efficiency Through Time



- Robust $Z^*(5)$ on windows of 500 days, one window every 21 days; dashed: ± 1.96
- No window rejects the random walk for Bitcoin
- Early studies found Bitcoin inefficient before 2013–2016 Urquhart (2016); markets became more efficient as they grew Tran & Leirvik (2020)

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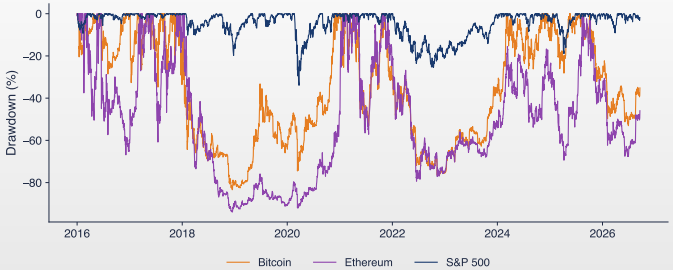
What Efficiency Does Not Mean

- Weak-form efficiency: the *direction* of returns is not predictable from past prices
  - it says nothing about whether the price is *right*
- A price can be unpredictable and still far from any fundamental value
  - Bitcoin has no cash flows: no dividends, no profits to discount
  - so "fundamental value" must come from use: payments, settlement, collateral, network size
- Two questions remain
  - how far can the price fall from a peak? (drawdowns)
  - can we detect a bubble, and does network activity justify the price? (log-periodic power law (LPPL), Metcalfe)

Drawdown: How Far from the Peak

- **Drawdown** on day t: $D_t = P_t / \max_{s \le t} P_s - 1$, the loss from the last record high
  - **maximum drawdown**: the deepest fall; **time under water**: the days until a new record
- A loss of d needs a gain of $d/(1-d)$ to recover
  - $d = 50\%$: $+100\%$; $d = 75\%$: $+300\%$; $d = 83.4\%$: $+502\%$
- Why managers care: a firm holding crypto on its balance sheet reports the drawdown, not the average return

Drawdowns of Bitcoin, Ethereum and the S&P 500



- Daily closes since 2016; maximum drawdowns: Bitcoin −83.4%, Ethereum −94.0%, S&P 500 −33.9%
- Bitcoin spent 76% of the days more than 10% below its record; the S&P 500 16%

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Bubbles and Crashes

- **Bubble**: a price far above any fundamental value, sustained by the expectation of selling higher
  - Cheah & Fry (2015): the price of Bitcoin shows speculative bubbles
- The idea of Johansen, Ledoit & Sornette (2000): imitation between traders creates **positive feedback**
  - prices grow *faster than exponentially*: the growth rate itself grows
  - the crash hazard rises as the price approaches a **critical time** t_c
  - the hierarchy of traders (individuals, funds, markets) adds oscillations that speed up near t_c
- The result is the **log-periodic power law** (LPPL), next slide



Didier Sornette, co-author of the LPPL model

Photo: Didier Sornette (2012); CC BY-SA 3.0 DE; Wikimedia Commons

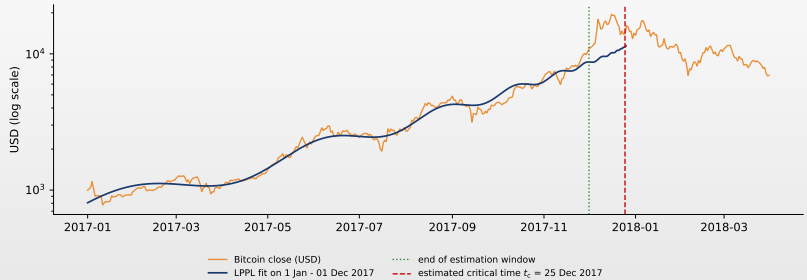
The LPPL Model

- Expected log price before the critical time, $t < t_c$:

$$\ln \mathrm{E}[P_t] = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos\big(\omega \ln(t_c - t) - \phi\big)$$

- Parameters
  - t_c: critical time, the most likely end of the bubble; A: the log price reached at t_c
  - $B < 0$ and $0 < m < 1$: super-exponential growth; $(t_c - t)^m$ shrinks to 0 as $t \to t_c$
  - C: amplitude of the oscillations; ω: their log-frequency; ϕ: their phase
- **Log-periodic**: the peaks of the cosine come at times t_k with $(t_c - t_{k+1})/(t_c - t_k) = e^{-2\pi/\omega}$
  - the gaps between peaks shrink geometrically: the oscillations accelerate towards t_c
- Conditions used to accept a fit as a bubble: $B < 0$, $0.1 \leq m \leq 0.9$, $6 \leq \omega \leq 13$, $|C| < 1$

Bitcoin 2017: an LPPL Fit



- Daily log close, window 1 January – 1 December 2017 ($n = 335$); two-step estimation, 300 random starts of the search over (t_c, m, ω)
- Estimated t_c = 25 December 2017; the actual peak came on 16 December 2017 (19 497 USD), followed by a fall of 83%

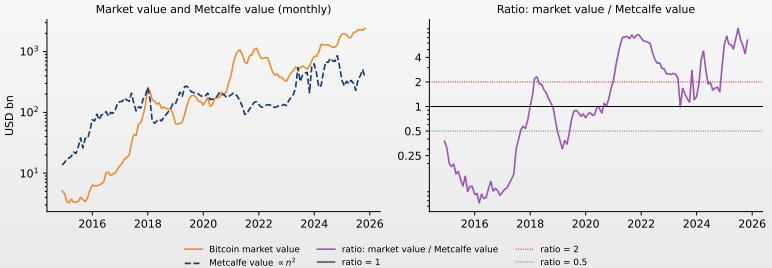
Network Value: Metcalfe's Law

- **Metcalfe's law**: the value of a network grows with the square of its users n
  - n users can form about $n^2/2$ connections; each connection adds value
  - used to value telecom and social networks; applied to Bitcoin by Peterson (2018)
- As a regression on logs:

$$\ln M_t = a + b \ln n_t + u_t, \qquad \text{Metcalfe: } b = 2$$

  - M_t: market value; n_t: a measure of use, here Bitcoin transactions per day; u_t: the error
  - b is an **elasticity**: 1% more use goes with b% more value
- Data: monthly means, November 2014 – October 2025, 132 months; the 11 bad days found in Lecture 1 removed

Market Value against Metcalfe Value



- Metcalfe value: $\exp(\hat{c} + 2 \ln n_t)$, with b fixed at 2 and $\hat{c}$ set so that the ratio is 1 on average in logs
- The ratio was below 0.5 in 2015–2016 and above 4 in 2021 and 2025 (maximum 9.1, June 2025)
- The ratio drifts upwards: value has grown faster than use; network size alone does not justify today's price

Summary

- Annualise with the market's own calendar: $\sigma_a = s\sqrt{365}$ gives 64.1% for Bitcoin, against 19.2% for the S&P 500
- Normality is rejected (Jarque–Bera $> 13\,000$ for all four assets); crash days are far more frequent than the Normal distribution allows
- Volatility clusters: Ljung–Box $Q(10) = 212.7$ for r_t^2, against 20.2 for r_t
- Since 2020 Bitcoin is a risk asset: correlation 0.39 with the S&P 500
- Robust variance-ratio tests do not reject weak-form efficiency, in any sub-period
- LPPL fits the 2017 bubble with $R^2 \approx 0.98$, but a good in-sample fit does not date the crash
- Metcalfe: $\hat{b} = 3.59$, standard error (SE) 0.43, $b = 2$ rejected; since 2020 $R^2 = 0.20$

Interpretation Exercises

1. A treasurer reports Bitcoin's annual volatility as 53.3%. What did she do wrong, and what is the right figure?
2. Jarque–Bera $= 29\,620$ for Bitcoin. State H_0, the conclusion, and the consequence for a VaR computed with the Normal distribution.
3. Bitcoin: Ljung–Box $Q(10) = 20.2$ (p 0.028) for r_t, 212.7 for r_t^2; robust $Z^*(5) = -0.19$ (p 0.853). Is Bitcoin predictable? In what sense?
4. LPPL fit on 1 January – 1 December 2017: $R^2 = 0.98$, $\hat{t}_c = 25$ December 2017. Would you have sold on 1 December? What does a high R^2 not tell you?
5. Metcalfe: $\hat{b} = 3.59$ (SE 0.43) on 2014–2025, $\hat{b} = 1.17$ with $R^2 = 0.20$ on 2020–2025. What does this imply for valuations based on user growth?

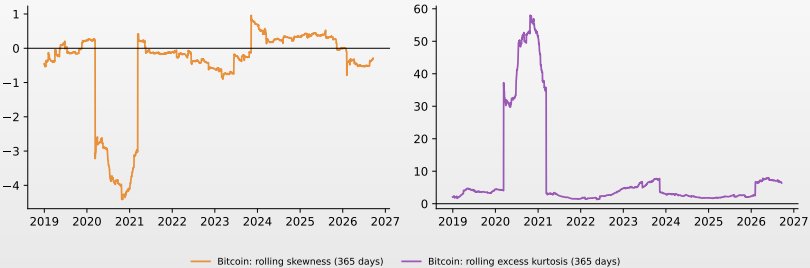
Watch Robust Markowitz for Crypto-Assets

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- Is Kurtosis a Stable Number?
- The Tail Index and the Hill Estimator
- Tail Indices of Crypto, Equities and Gold
- Tail Plot and Hill Plot
- A 7-Day Market with a Weekend Lull
- Three Crisis Episodes
- Hedge, Safe Haven or Risk Asset?
- Returns on the Worst Equity Days
- Estimating the LPPL
- LPPL Estimates: How Stable Is $\hat{t}_c$?
- $\hat{t}_c$ Window by Window
- Metcalfe's Law: the Estimates
- A Rolling Metcalfe Elasticity
- Measuring Volatility Through Time
- Rolling Correlation
- The Robust Variance-Ratio Statistic
- Worked Example: Variance Ratio from Autocorrelations
- The Variance-Ratio Profile

Is Kurtosis a Stable Number?



- Skewness and excess kurtosis of Bitcoin on rolling 365-day windows
- Kurtosis ranges from 1.3 to 58.0 (window ending October 2020, which contains 12 March 2020); today 6.4
- One day moves the estimate by a factor of ten: with $\alpha < 4$ the sample kurtosis has no stable limit (previous slides)

The Tail Index and the Hill Estimator

- Far in the tail, losses follow a power law:

$$P(L > x) \approx c\, x^{-\alpha} \quad \text{for large } x$$

  - $L = -r$: the daily loss; $c > 0$: a constant; α: the **tail index**
  - smaller α = heavier tail; moments of order $q \geq \alpha$ are infinite: $\alpha < 4$ means infinite kurtosis
- Hill (1975) estimator from the k largest losses $L_{(1)} \geq L_{(2)} \geq \cdots \geq L_{(k+1)}$, with a confidence interval (CI):

$$\hat{\alpha} = \left[\frac{1}{k} \sum_{i=1}^{k} \ln \frac{L_{(i)}}{L_{(k+1)}} \right]^{-1}, \qquad \mathrm{CI}_{95\%} = \hat{\alpha}\left(1 \pm \frac{1.96}{\sqrt{k}}\right)$$

  - $L_{(k+1)}$: the threshold of the tail; here $k = 2.5\%$ of n
  - large ratios $L_{(i)}/L_{(k+1)}$ on average give a small $\hat{\alpha}$: a heavy tail

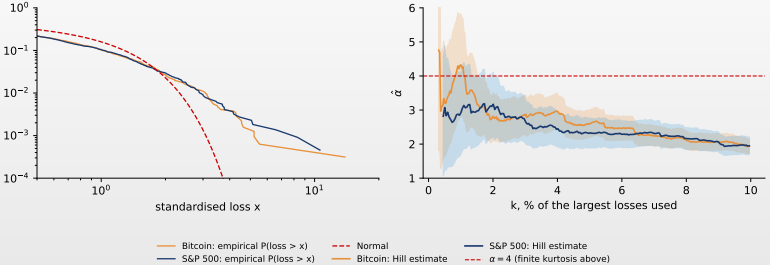
Tail Indices of Crypto, Equities and Gold

	left tail $\hat{\alpha}$	95% CI	right tail $\hat{\alpha}$	95% CI
Bitcoin	2.71	[2.20, 3.21]	3.37	[2.74, 4.00]
Ethereum	2.62	[2.10, 3.15]	3.09	[2.48, 3.71]
S&P 500	2.68	[2.08, 3.29]	2.80	[2.17, 3.43]
Gold	3.11	[2.42, 3.80]	3.04	[2.36, 3.71]

- Daily returns since 2014 (Bitcoin, S&P 500, gold) and 2016 (Ethereum), to 18 September 2026
- Interpretation
  - all left-tail indices are below 4: the sample kurtosis of the previous table is not a stable number
  - crypto is more *volatile*, but the *shape* of its tails is close to that of equities; the intervals overlap
  - for Bitcoin, losses have a heavier tail than gains (2.71 against 3.37)

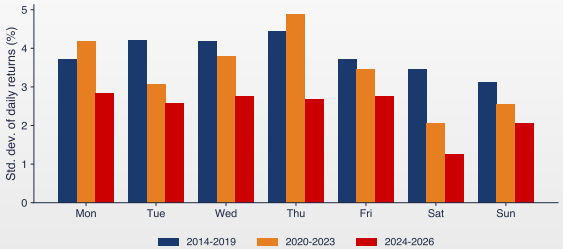
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Tail Plot and Hill Plot



- Left: $P(\text{loss} > x)$ on log-log scales; a power-law tail is a straight line with slope $-\alpha$, the Normal (red) bends down
- Right: Hill estimate against k with 95% bands, 2018–2026. Bitcoin: 4.33 ($k = 1\%$), 2.71 (2.5%), 2.60 (5%), 1.94 (10%); S&P 500: 2.92, 2.81, 2.32, 1.95
- Small k: few observations, wide bands; large k: the centre of the distribution biases $\hat{\alpha}$ down. Report $\hat{\alpha}$ on the flat part, here $k \approx 2.5\%$

A 7-Day Market with a Weekend Lull



- Standard deviation of Bitcoin's daily returns by closing day, three periods
- No weekend effect in the *mean*; but weekend variance is only 35–66% of weekday variance
- Crypto trades on Saturdays, but banks, macro news and institutional traders do not: risk follows the traditional calendar

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Three Crisis Episodes

Episode	period	Bitcoin	Ethereum	S&P 500	gold
COVID-19	19 Feb – 23 Mar 2020	−40.6	−65.5	−41.4	0.3
Terra/Luna	5 – 12 May 2022	−23.0	−33.8	−5.4	−3.0
FTX	5 – 21 Nov 2022	−29.9	−38.4	4.6	3.4

- Cumulative log returns in % over each episode
- Interpretation
  - a global shock (COVID-19) hits crypto as hard as equities; gold does not move
  - crypto-specific shocks (the collapse of the stablecoin TerraUSD; the bankruptcy of the exchange FTX) hit crypto by a quarter or more, equities hardly at all
  - the link is **asymmetric**: equity crashes reach crypto, crypto crashes rarely reach equities

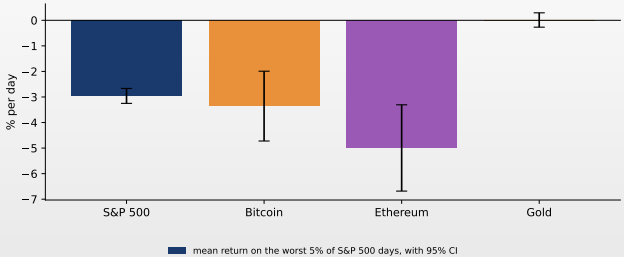
Hedge, Safe Haven or Risk Asset?

- Baur & Lucey (2010): a **hedge** is uncorrelated or negatively correlated with equities on average; a **safe haven** keeps its value when equities crash
- Test: the mean return of asset x on the set $\mathcal{S}$ of the worst 5% of S&P 500 days

$$\bar{x}_{\mathcal{S}} = \frac{1}{n_{\mathcal{S}}} \sum_{t \in \mathcal{S}} x_t, \qquad t = \frac{\bar{x}_{\mathcal{S}}}{s_{\mathcal{S}} / \sqrt{n_{\mathcal{S}}}}$$

  - 2018–2026, common days: $n_{\mathcal{S}} = 110$ days with the S&P 500 below -1.77%, on average -2.96%
  - safe haven: $\bar{x}_{\mathcal{S}} \geq 0$
- Results: Bitcoin -3.36% ($t = -4.82$), Ethereum -4.99% ($t = -5.80$), gold $+0.01\%$ ($t = 0.09$)
- Verdict: crypto falls *more* than equities on the worst equity days; gold is a safe haven

Returns on the Worst Equity Days



- Mean daily return on the 110 worst S&P 500 days of 2018–2026, with 95% confidence intervals
- For a corporate treasury: Bitcoin adds crash risk exactly when the rest of the balance sheet is under stress

Estimating the LPPL

- Rewrite the cosine term with $C_1 = C\cos\phi$ and $C_2 = C\sin\phi$:

$$\ln P_t = A + Bf_t + C_1 g_t + C_2 h_t + \varepsilon_t$$

$$f_t = (t_c - t)^m, \quad g_t = (t_c - t)^m \cos(\omega \ln(t_c - t)), \quad h_t = (t_c - t)^m \sin(\omega \ln(t_c - t))$$

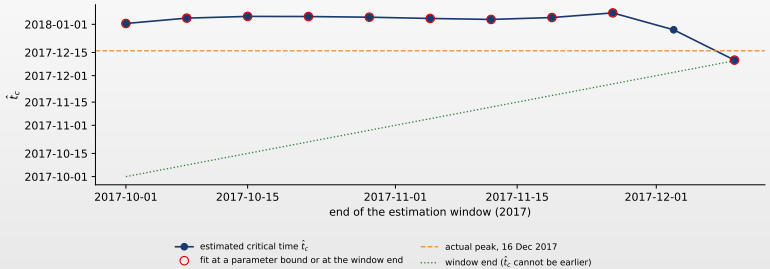
- Two steps
  1. for given (t_c, m, ω), the model is linear in (A, B, C_1, C_2): ordinary least squares
  2. search over (t_c, m, ω) to minimise the sum of squared errors $\sum_t \varepsilon_t^2$
- Caution
  - many local minima: start the search from many points; fit on many windows
  - t_c is the least precise parameter; fits look convincing after the crash, real-time warnings give false alarms

LPPL Estimates: How Stable Is $\hat{t}_c$?

window ends	n	$\hat{t}_c$	lead (days)	$\hat{m}$	$\hat{\omega}$	$\lvert\hat{C}\rvert/\lvert\hat{B}\rvert$	R^2
15 Nov 2017	319	4 Jan 2018	50	0.900	13.00	0.089	0.980
1 Dec 2017	335	25 Dec 2017	24	0.851	12.49	0.088	0.982
8 Dec 2017	342	8 Dec 2017	0	0.741	11.14	0.081	0.979

- All windows start on 1 January 2017; $\hat{B} \approx -2.8$ in all three: super-exponential growth
- Interpretation
  - $R^2 \approx 0.98$ in every window: a good fit does not mean a reliable forecast
  - $\hat{t}_c$ moves by four weeks when the window moves by three weeks
  - 15 November: $\hat{m}$ and $\hat{\omega}$ sit on the bounds of the search (0.9 and 13), a sign of a poorly identified fit
  - 8 December: $\hat{t}_c$ equals the last observation, a degenerate solution
- Practice: fit many windows and report the share of windows that pass the filters (the log-periodic power law singularity (LPPLS) confidence indicator)

$\hat{t}_c$ Window by Window



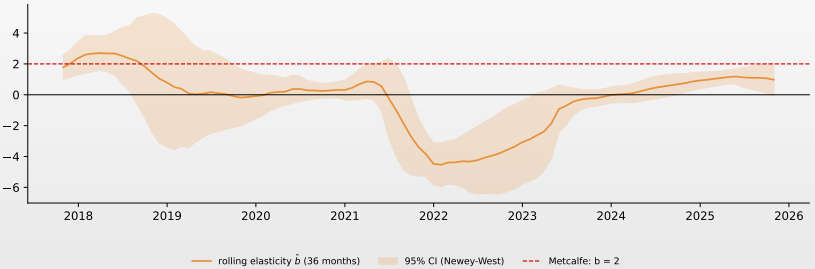
- Eleven weekly window ends, 1 October – 10 December 2017, all starting 1 January 2017; 120 random starts each
- Nine of eleven fits sit at the bounds $m = 0.9$, $\omega = 13$ (red circles), with $\hat{t}_c$ between 1 and 7 January 2018
- Only the window ending 3 December gives an interior fit ($\hat{t}_c$ = 28 December); the 10 December fit collapses onto the window end

Metcalfe's Law: the Estimates

Sample	months	$\hat{b}$	standard error	R^2
2014–2025	132	3.59	0.43	0.58
2014–2019	62	2.71	0.35	0.52
2020–2025	70	1.17	0.48	0.20

- Standard errors robust to autocorrelation (Newey–West, 12 lags)
- Test of $b = 2$ on the full sample: $t = (3.59 - 2)/0.43 = 3.66$; rejected at 5%
- Interpretation
  - until 2019, value and use moved together, with an elasticity above 2
  - since 2020 the link is weak ($R^2 = 0.20$): ETFs, institutional investors and macro news move the price, not transactions
  - transactions are an imperfect proxy for users: one firm can send many transactions, and many users trade off-chain on exchanges

A Rolling Metcalfe Elasticity



- $\hat{b}$ on rolling 36-month windows with Newey–West 95% intervals; red: Metcalfe's $b = 2$
- $\hat{b}$ peaks at 2.70 (window ending March 2018), turns negative in 2021–2023 (value fell while transactions grew), ends at 0.98 (October 2025)
- The interval covers 2 in only 28% of the windows: there is no stable network-value elasticity

Measuring Volatility Through Time

- **Rolling volatility** on a window of the last w days, annualised:

$$\hat{\sigma}_{a,t} = \sqrt{A}\sqrt{\frac{1}{w-1}\sum_{j=0}^{w-1}\left(r_{t-j} - \bar{r}_t\right)^2}, \qquad \bar{r}_t = \frac{1}{w}\sum_{j=0}^{w-1} r_{t-j}$$

  - w: the window length (30 days for crypto, 21 trading days for equities, about one month)
  - A: returns per year (365 or 252); the window moves forward one day at a time
- Trade-off in the choice of w
  - short window: reacts fast, but noisy; long window: smooth, but late
  - the standard error of $\hat{\sigma}$ is roughly $\hat{\sigma}/\sqrt{2w}$: about 13% of $\hat{\sigma}$ for $w = 30$
- Lecture 3 replaces the window by a model: exponentially weighted moving average (EWMA) or GARCH

Rolling Correlation

- Pearson correlation of two return series x (Bitcoin) and y (S&P 500) on the last w common days:

$$\hat{\rho}_{xy,t} = \frac{\sum_{j=0}^{w-1}(x_{t-j} - \bar{x}_t)(y_{t-j} - \bar{y}_t)}{\sqrt{\sum_{j=0}^{w-1}(x_{t-j} - \bar{x}_t)^2}\sqrt{\sum_{j=0}^{w-1}(y_{t-j} - \bar{y}_t)^2}}$$

  - $w = 250$ common days, about one year of equity trading
  - standard error $\approx (1 - \rho^2)/\sqrt{w} \approx 0.06$: changes below 0.1 are noise
- Two calendars
  - join *prices* on common days first, then compute returns: Monday's Bitcoin return then covers Friday to Monday, like the equity return
  - closes are not simultaneous (midnight UTC, Coordinated Universal Time, vs 16:00 New York): weekly returns give a check

The Robust Variance-Ratio Statistic

- Volatility clustering makes the independent and identically distributed (i.i.d.) test too eager to reject; Lo & MacKinlay (1988) give a robust version:

$$Z^*(q) = \frac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{\hat{\theta}^*(q)}} \;\approx\; N(0,1)$$

$$\hat{\theta}^*(q) = \sum_{j=1}^{q-1} \left[\frac{2(q-j)}{q}\right]^2 \hat{\delta}_j, \qquad \hat{\delta}_j = \frac{\sum_{t=j+1}^{n} (r_t - \bar{r})^2 (r_{t-j} - \bar{r})^2}{\left[\sum_{t=1}^{n} (r_t - \bar{r})^2\right]^2}$$

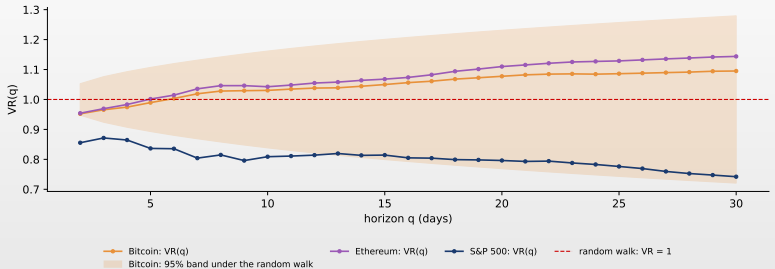
  - $\hat{\delta}_j$: the variance of the lag-j autocorrelation when volatility changes over time
  - reject the random walk at 5% if $|Z^*(q)| > 1.96$

- Link with autocorrelations: $\mathrm{VR}(q) \approx 1 + 2\sum_{j=1}^{q-1} \left(1 - \frac{j}{q}\right)\rho(j)$
  - positive autocorrelations push VR above 1, negative ones below 1

Worked Example: Variance Ratio from Autocorrelations

- For $q = 2$: $\mathrm{VR}(2) \approx 1 + 2\,(1 - \tfrac{1}{2})\,\rho(1) = 1 + \rho(1)$
- Bitcoin, since 2014: $\hat{\rho}(1) = -0.022$, so $\mathrm{VR}(2) \approx 0.978$; the direct estimate is 0.977
- S&P 500, same period: $\hat{\rho}(1) = -0.126$, so $\mathrm{VR}(2) \approx 0.874$; the direct estimate is 0.875
- Interpretation
  - the S&P 500 index shows mild daily mean reversion, typical of an index (bid-ask bounce, stale prices of its members)
  - the robust test does not reject because the variance of $\widehat{\mathrm{VR}}$ is large when volatility clusters

The Variance-Ratio Profile



- $\widehat{\mathrm{VR}}(q)$ for $q = 2, \ldots, 30$ days, 2018–2026; shaded: the robust 95% band for Bitcoin
- Bitcoin: VR(5) = 0.989 ($Z^* = -0.19$), VR(20) = 1.077 ($Z^* = 0.66$); Ethereum VR(20) = 1.11 ($Z^* = 0.95$)
- S&P 500: VR(20) = 0.796 ($Z^* = -0.76$): mean reversion in the point estimate, but not significant

Bitcoin's Large Drawdowns

Peak	USD	trough	USD	depth (%)	new record	days under water
2014-09-17	457	2015-01-14	178	−61.1	2015-12-15	454
2017-12-16	19 497	2018-12-15	3 237	−83.4	2020-11-30	1 080
2021-04-13	63 503	2021-07-20	29 807	−53.1	2021-10-19	189
2021-11-08	67 567	2022-11-21	15 787	−76.6	2024-03-04	847
2025-10-06	124 753	2026-06-30	58 559	−53.1	not yet	–

- Episodes since September 2014 deeper than 50%
- Interpretation
  - five falls of more than half in twelve years: a crash of this size is a regular event
  - the 2017 and 2021 peaks took 1 080 and 847 days to recover

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Worked Example: Accelerating Oscillations

- Peaks of the cosine term satisfy $\omega \ln(t_c - t_k) - \phi = 2\pi k$, so consecutive distances to t_c shrink by a constant factor

$$\frac{t_c - t_{k+1}}{t_c - t_k} = e^{-2\pi/\omega}$$

- Bitcoin 2017 fit (next slides): $\hat{\omega} = 12.49$, so $e^{-2\pi/12.49} = 0.605$
  - a peak 30 days before t_c is followed by peaks 18.1, 11.0 and 6.6 days before t_c
- The ratio $\lambda = e^{2\pi/\omega} = 1.65$ is the "preferred scaling ratio" of the hierarchy of traders in Johansen, Ledoit & Sornette (2000)
- Practical reading: shrinking gaps between local tops are the visual signature of an LPPL bubble